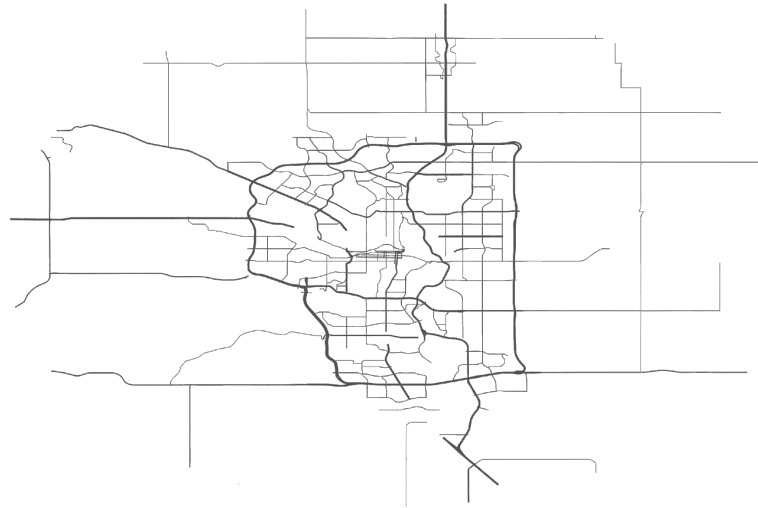


ANALYST PORTAFOLIO



ADRIAN CHUQUIPIONDO



ABOUT ME

+1 647 787 2946

adrianchuquipiondo@gmail.com

Architect with vast experience as a researcher and consultant in Urban Development and Real Estate market analysis projects.

Coordinated +10 research and consultancy projects in Peru, Mexico, and India.

Analyses statistical and spatial data, creating visualizations and maps to inform decision making.

GIS ANALYSIS

INSIGHTS AND RESULTS

This is a summary analysis of the **Auto Theft Open Data** (*Toronto Police Service Open Data Portal*).

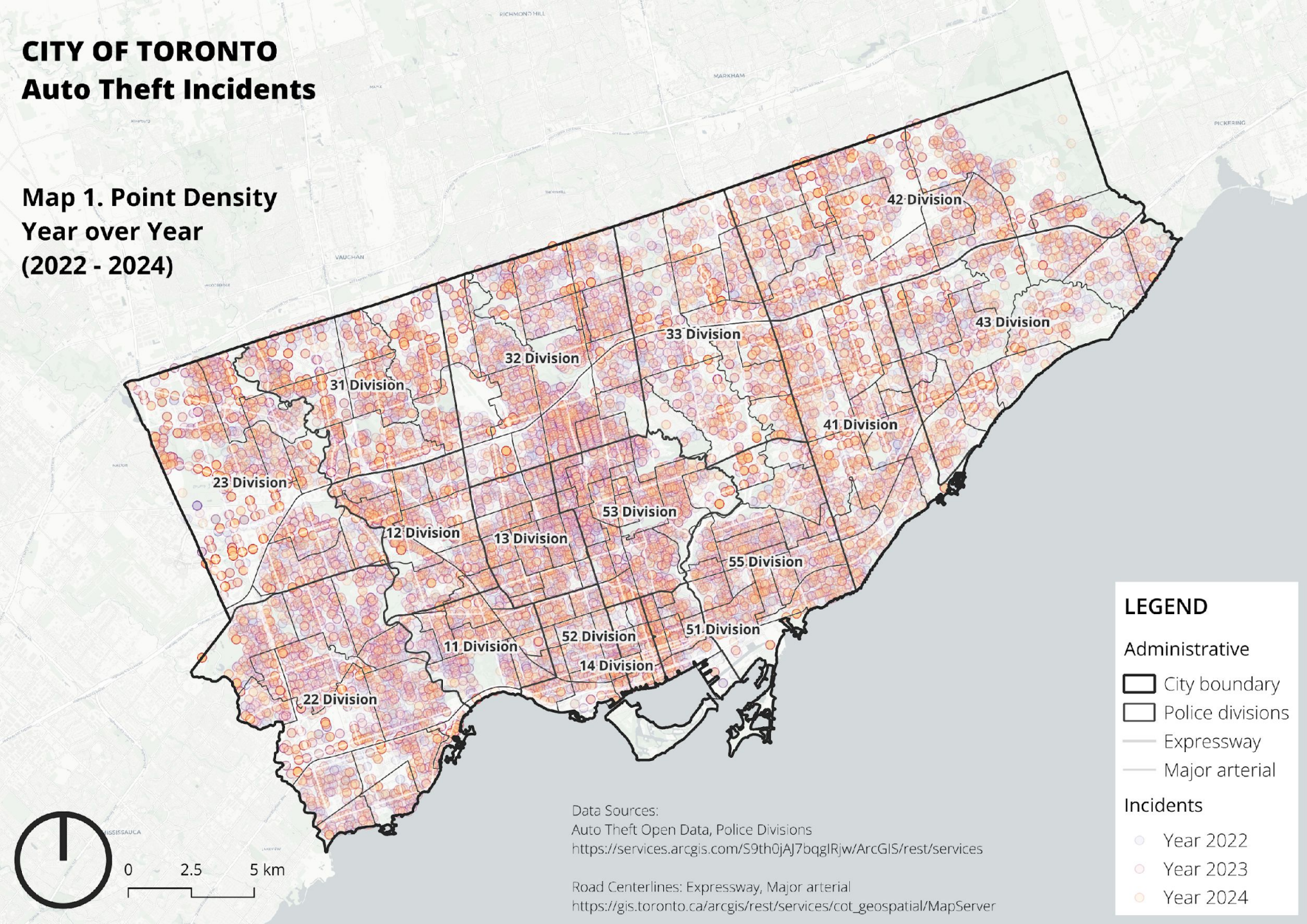
First published on March 28, 2023. The current version was updated on June 6, 2025. It has 70 163 records and 28 fields.

This analysis focuses on the time frame of **2022 - 2024**. Which will be useful to visualize trends since the end of covid-19.

Data from 2025 is not considered as the year is still in progress and does not yet provide a complete picture for comparison.

CITY OF TORONTO Auto Theft Incidents

Map 1. Point Density
Year over Year
(2022 - 2024)



INSIGHTS AND RESULTS

The Toronto covid-19 quarantine ended in May 2022. Before 2022, the average number of auto theft was 1, 142.

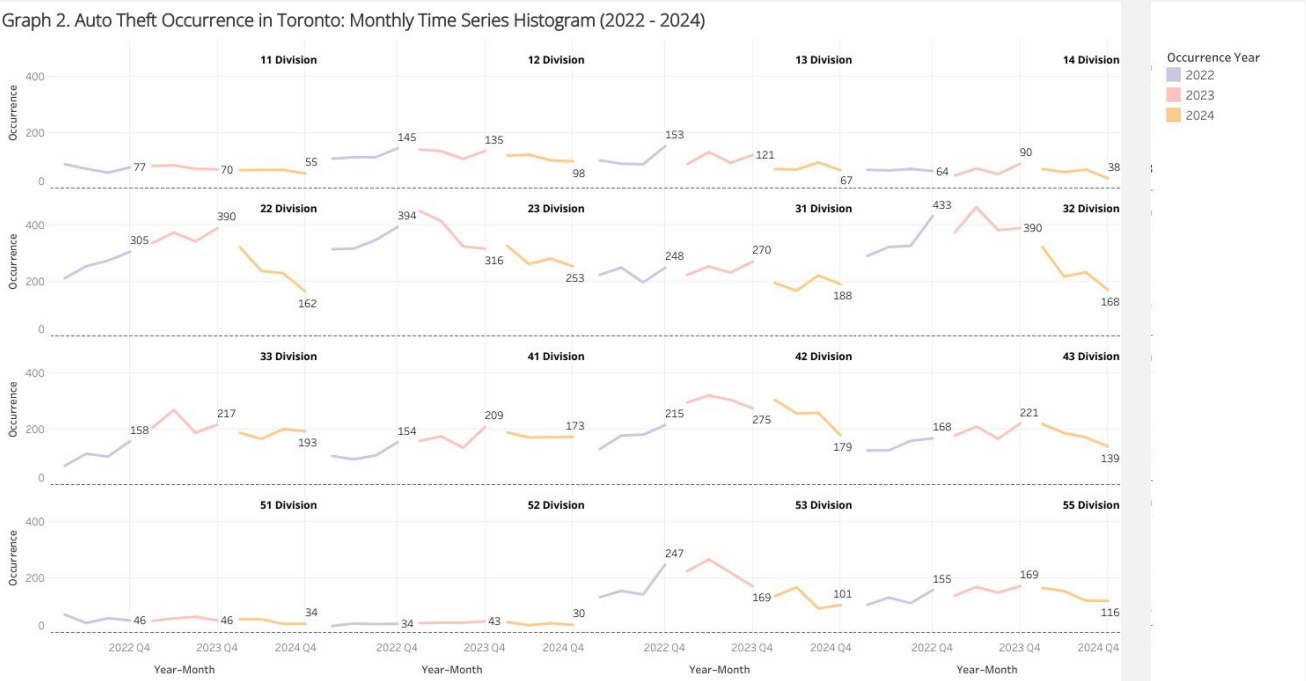
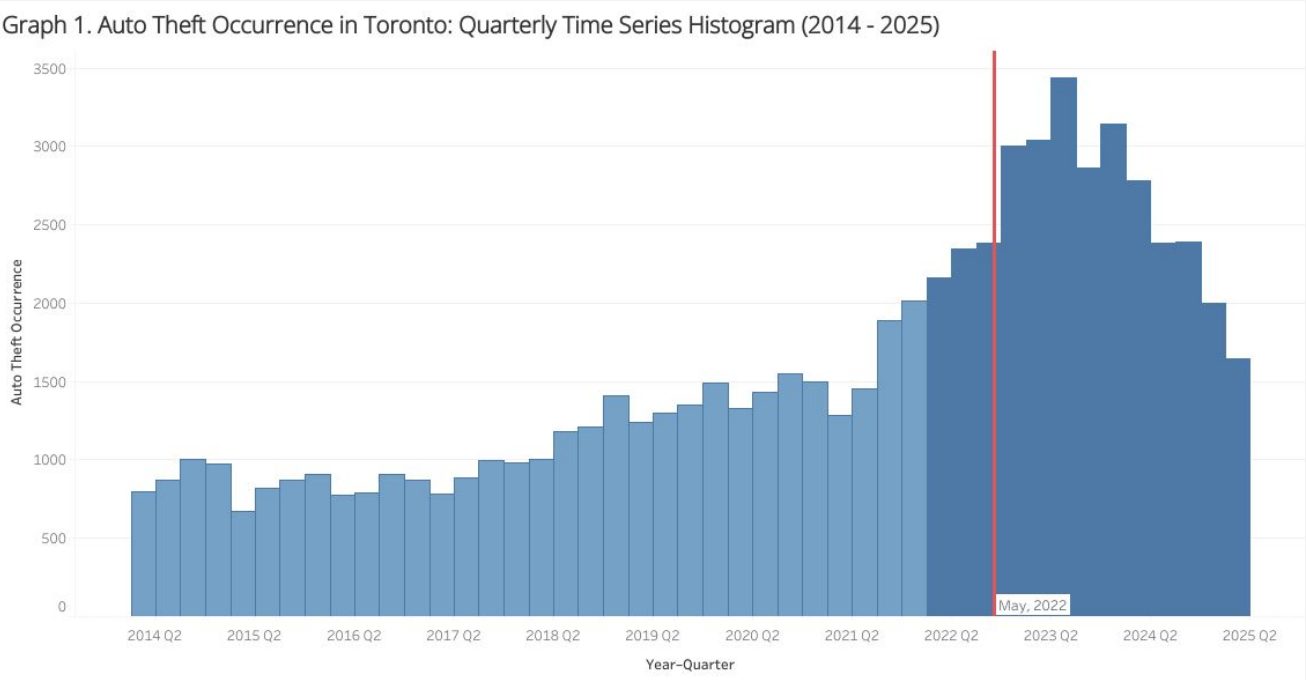
Between January and May 2022, the average increased to 2, 295. After the end of the quarantine, it remained high at 2, 269.

The *Graph 1* depicts the stepped increase after covid-19, and shows that the number incidents reached a peak in Q2 of 2023.

In recent years, the data resembles a normal distribution, meaning that after the peak in Q2 of 2023, the number of incident will likely decrease.

A disaggregation by Police Divisions shows this trend is quite consistent.

The *Graph 2* indicates that in most cases, Q4 of 2024 had fewer incidents than Q4 of 2022. Only in the cases of Division 33 and 41 did the number of incidents increase.



INSIGHTS AND RESULTS

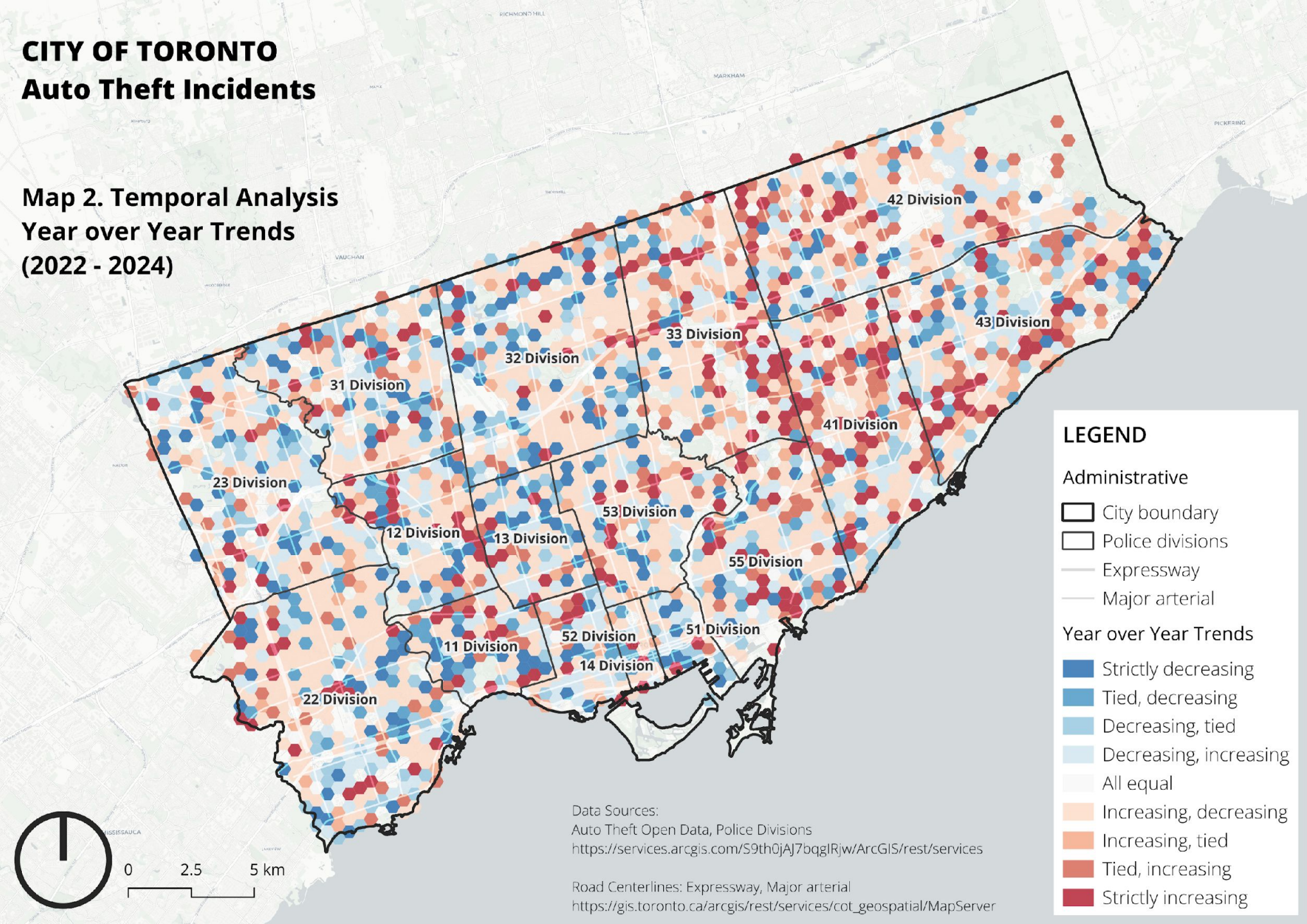
A detailed look at City of Toronto, divided into a hexagonal grid, shows trends over recent years. Specifically, in areas where values increased, decreased, remained the same, or fluctuated.

The *Map 2* shows that the spatial patterns of the bins that strictly increased or decreased are scattered across the city.

In contrast, the bins that increased and then decreased tend to form clusters. For example, in the Divisions 22, 12, 53, 32 and 33.

CITY OF TORONTO Auto Theft Incidents

Map 2. Temporal Analysis
Year over Year Trends
(2022 - 2024)



INSIGHTS AND RESULTS

Divisions 12, 22, 32, 33, and 52 do not contain all of the largest areas, nor are their auto theft occurrences particularly higher than those of other divisions.

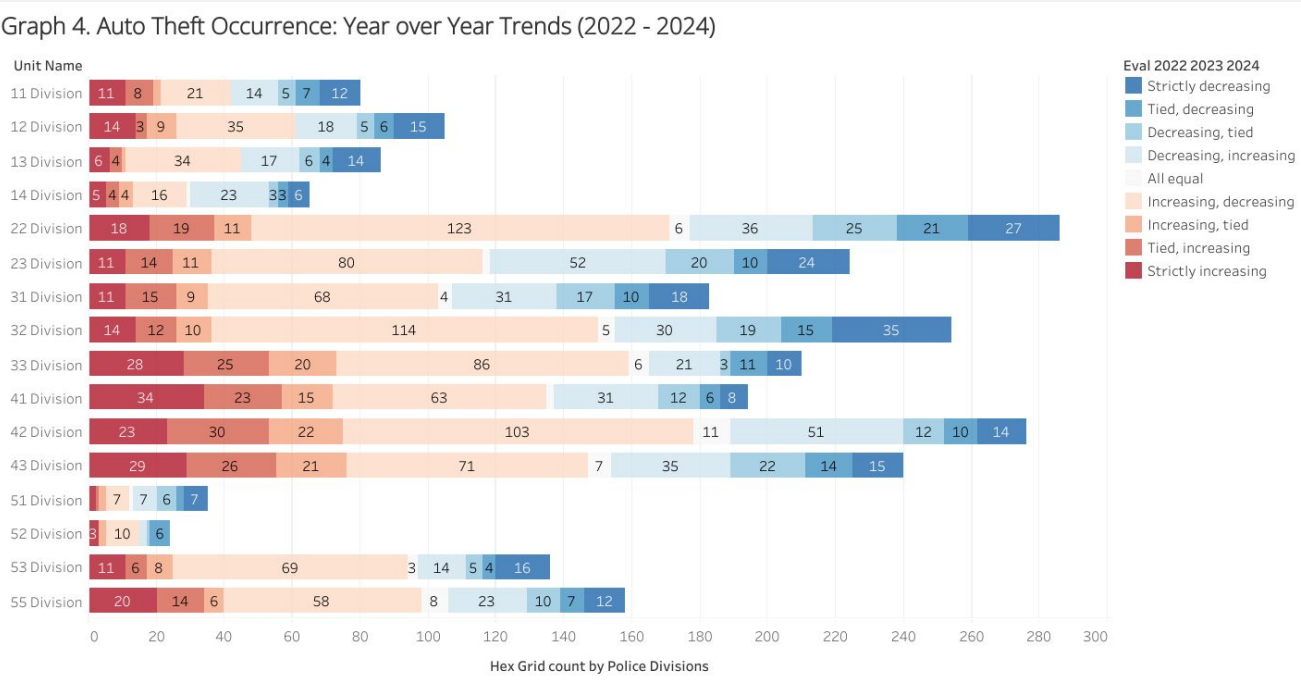
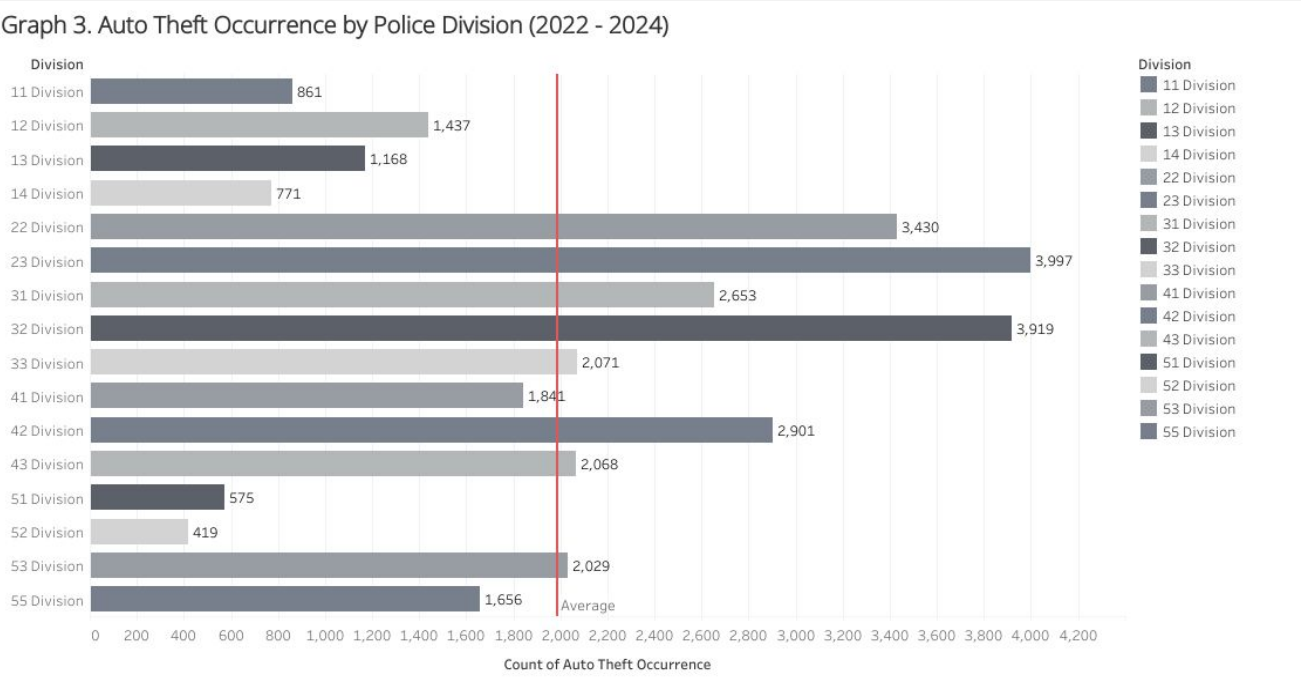
The *Graph 3* indicates that their values are distributed both above and below the average of 1,987 incidents.

It is also notable that Divisions 22 and 32 have larger areas and the highest number of incidents.

The *Graph 4* shows that occurrences at the division level did not remain consistent between 2022 and 2024.

The ratio between divisions with **strictly decreasing** trends and **strictly increasing** is quite similar, slightly favoring of the strictly decreasing ones.

Similarly, when comparing divisions that **increased then decreased** to those that **decreased then increased**, the former is more predominant.



INSIGHTS AND RESULTS

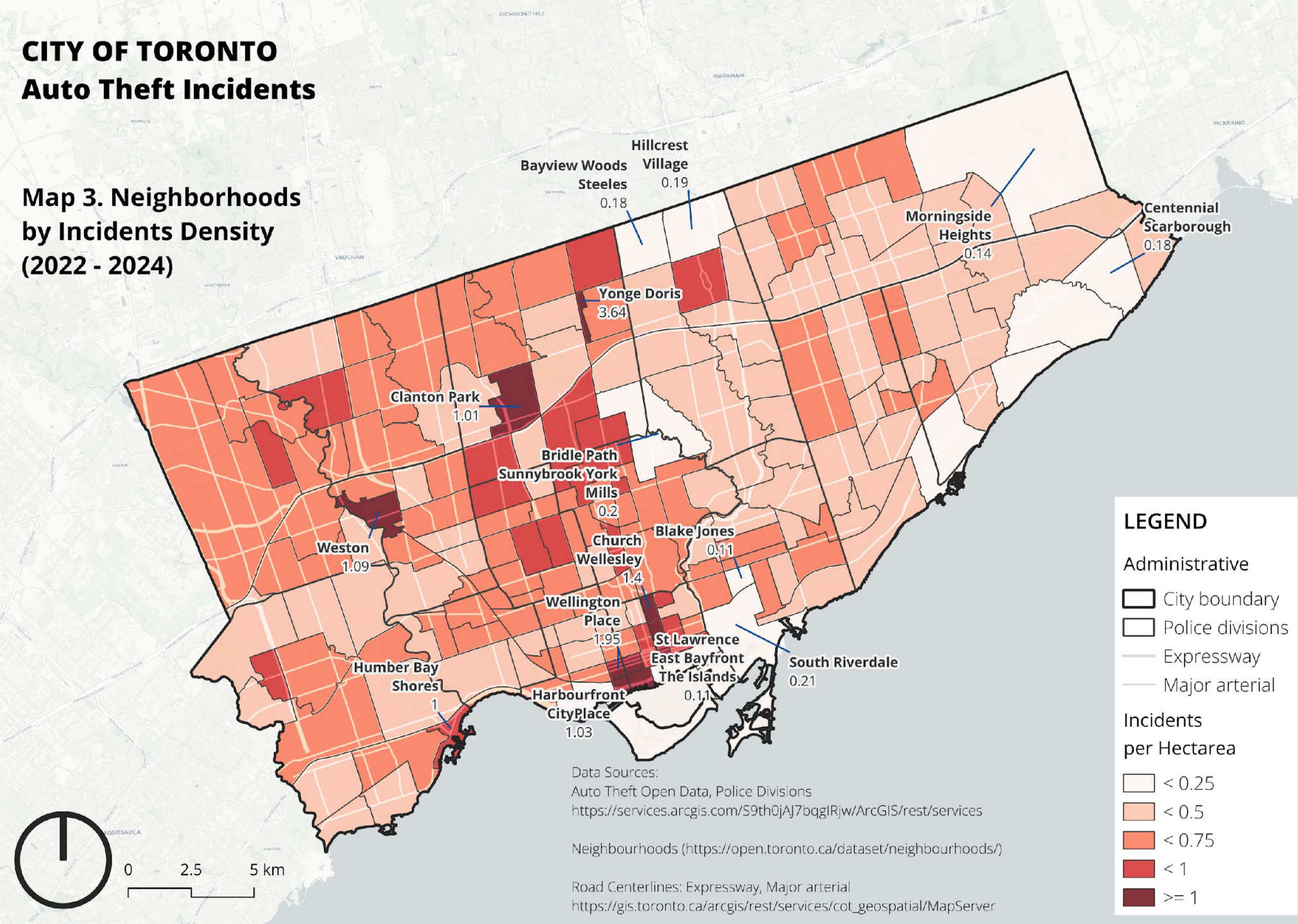
In order to avoid biased conclusions, the auto theft incidents need to be normalized. Using the neighborhoods areas shows a different pattern.

There are seven neighborhoods with more than one auto theft incidents per hectare, located within the Divisions 14, 12, 32, 51, and 52.

This is significant, as having more than one incident per hectare is approximately one standard deviation above the mean.

CITY OF TORONTO
Auto Theft Incidents

Map 3. Neighborhoods by Incidents Density (2022 - 2024)



INSIGHTS AND RESULTS

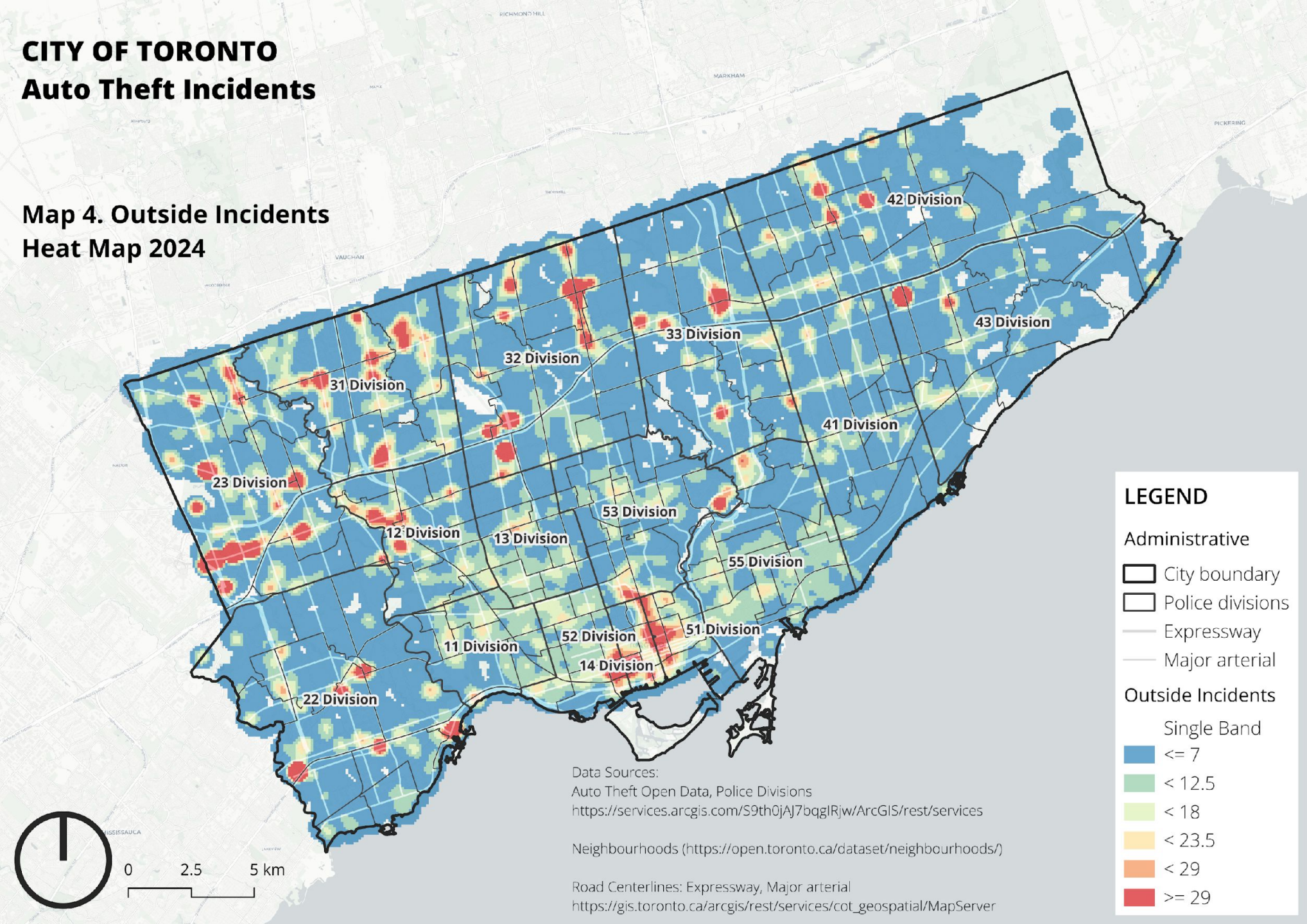
A detailed analysis of outside Incidents in 2024 reveals that very high hotspots are mainly located between neighborhoods and along major arterial roads.

This is particularly evident in Divisions 14, 23, 32, and 41.

There is a patch of moderate hotspots in Division 11, 13, 14 and 52, with another one in Division 55.

CITY OF TORONTO Auto Theft Incidents

Map 4. Outside Incidents
Heat Map 2024



REAL ESTATE

RESIDENTIAL MARKET ANALYSIS

The following is sample information of a monthly report of the apartment market in one of the most high-demand areas in the city of Lima.

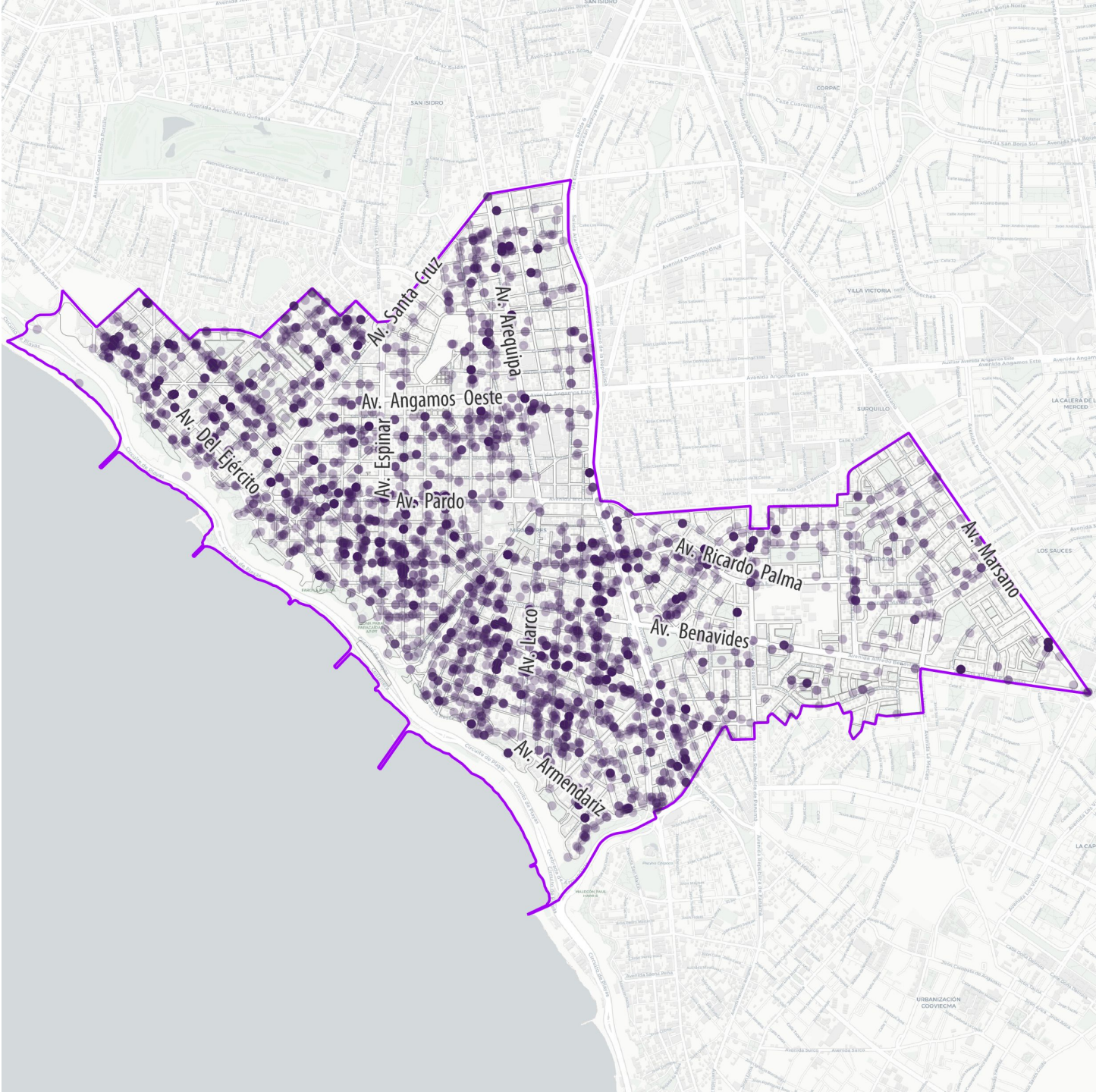
It was developed by Real Estate Analytics SAC. An start-up I co-founded. It's main goal was to inform business decisions.

As a part of the start-up, my role was to analyze large volumes of monthly data, clean, process and design visualizations. As well as descriptive statistics.

SALES (6 773 units)
Average Size 102 square meters
Average Price 8 595 soles/m2

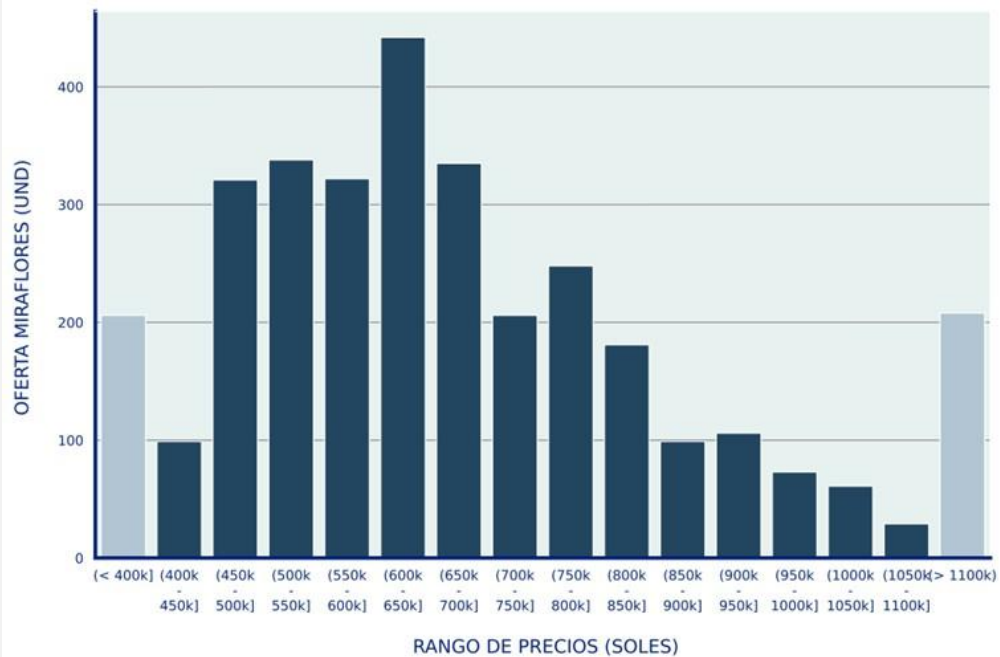
FOR RENT (1 247 units)
Average Size 107 square meters
Average Price 36 soles/m2

PERFORMANCE
RENT vs SALES 5.4x
Price-to-Earnings 21



DESCRIPTIVE
STATISTICS

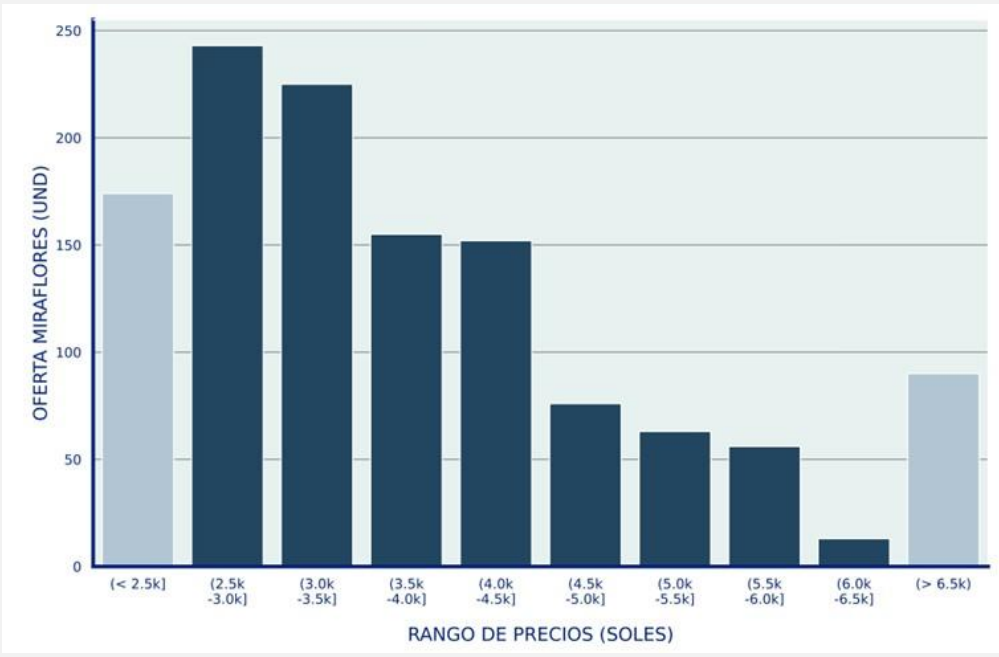
Histogram New Apartment units vs Price



The largest supply of new housing is in the price range between S/ 600k – 650k soles. In contrast, the lowest supply is found in the range between S/ 1 050k – 1 100k soles.

The **average** is S/ 687 749 soles. The **median** S/ 731 371 soles. When comparing the median to the average, the **ratio of 1 to 1.06** indicates that most of the supply is concentrated near the average.

Histogram For Rent units vs Price



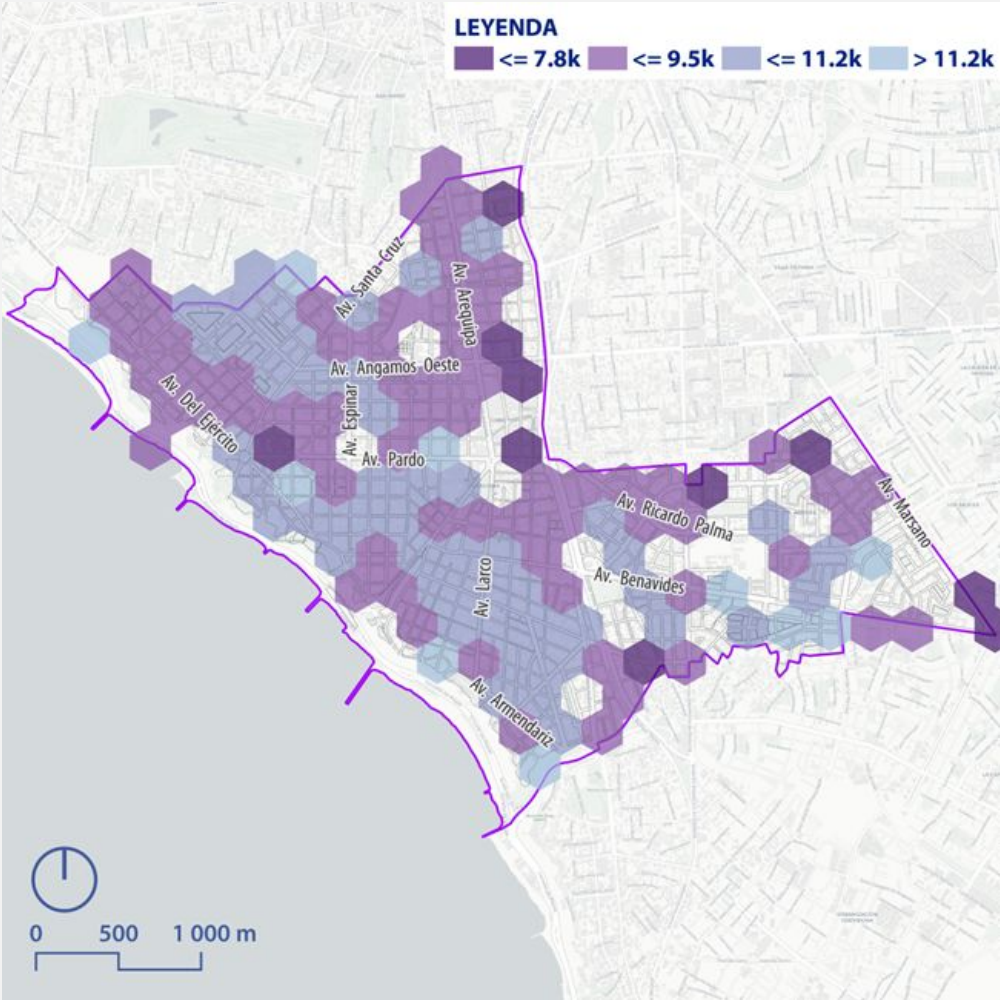
The largest supply of rental housing is in the price range between S/ 2 500 and 3 000 soles. In contrast, the lowest supply is found in the range between S/ 6,000 and 6,500 soles.

The **average** is S/ 3 869 soles. The median rental price is S/ 3 368 soles. The median to the average has a **ratio of 1 to 0.87**, indicating that most of the supply is concentrated around the average price.

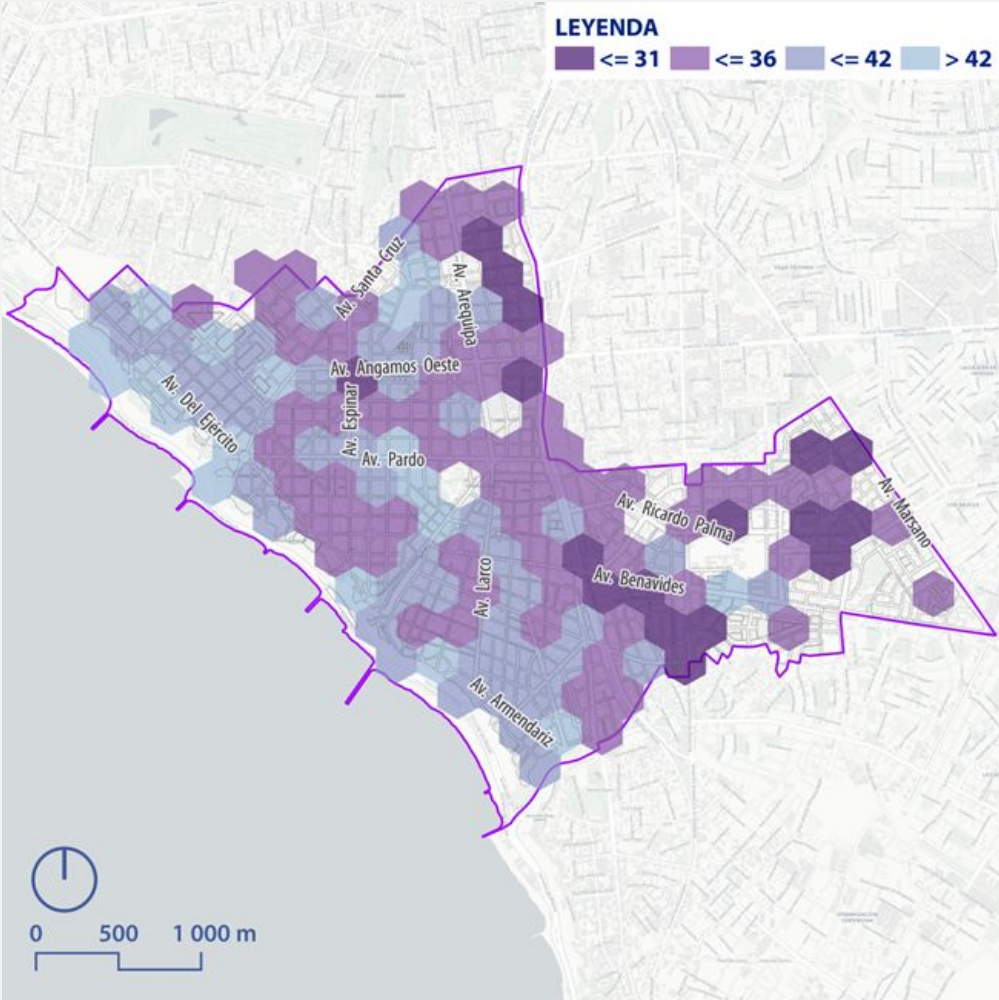
**SPATIAL PATTERNS
AGGREGATED
DATA**

The report included maps to allow users to locate and compare housing supply by area and price. The price ranges were derived from mean and standard deviation values.

It also included maps for performance metrics, such as Sales vs Rent and Price per Earnings ratio.



Sale price for New Residential units



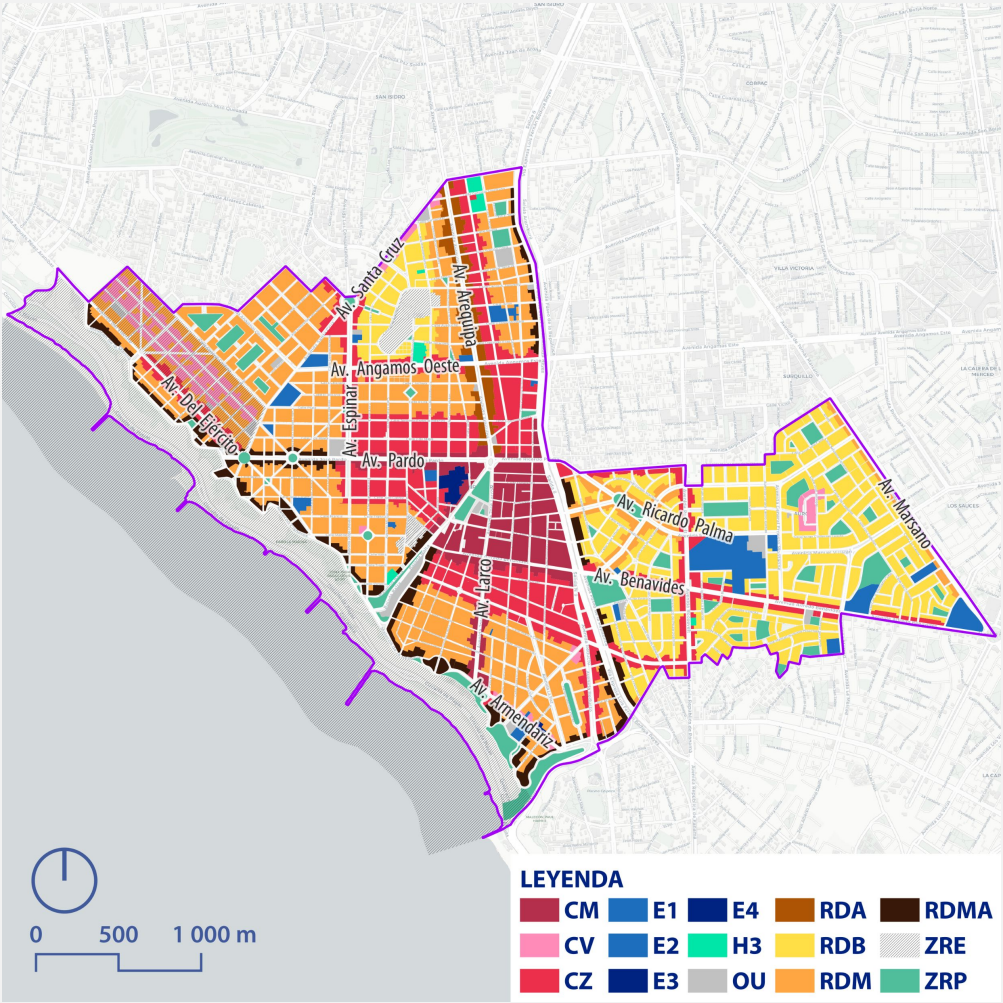
For Rent price for Residential units

ZONIN BY LAW & SOCIO-ECONOMICS

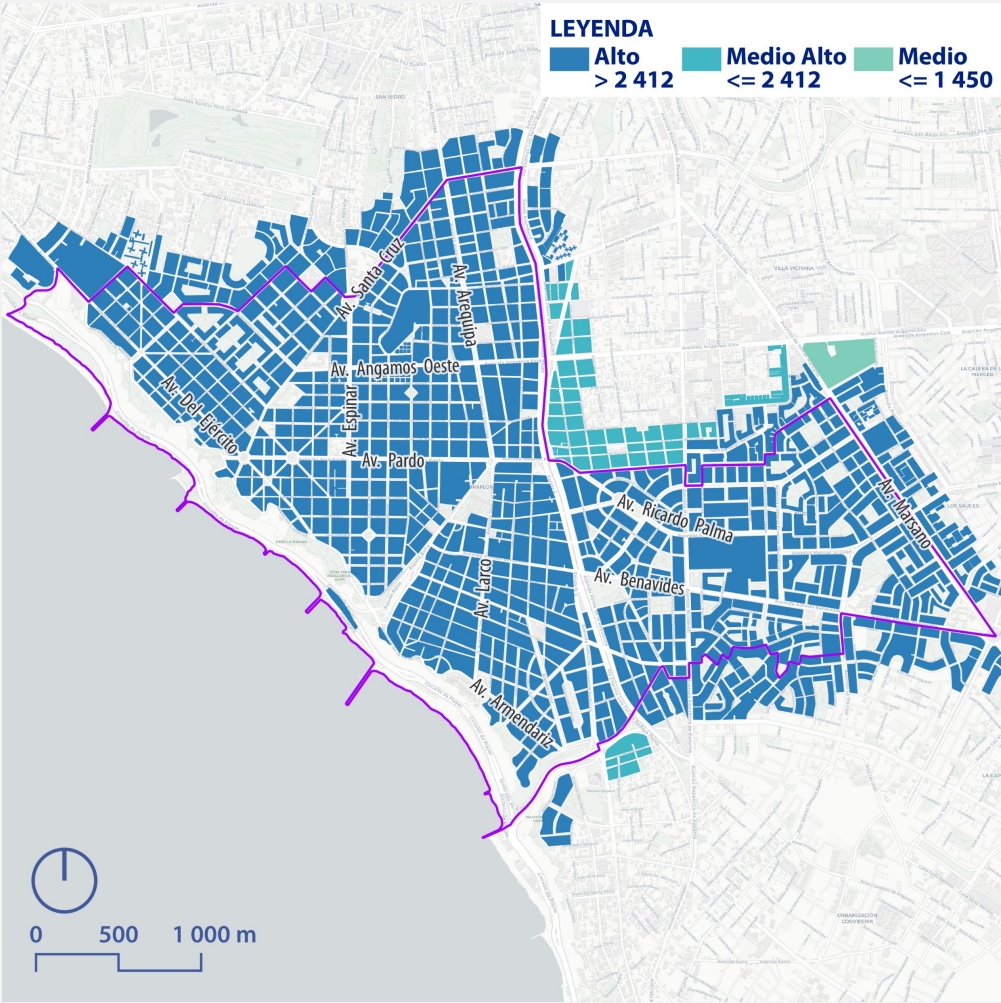
Complementary information was also provided to allow users to consider the most up-to-date data from trustworthy sources.

The Zonin by Land Use map is a public resource *provided by the District Municipality*.

The socioeconomic level (NSE) map is based on the 2020 *Stratified Map of Metropolitan Lima*. It illustrates the distribution of formal per capita income within the district.

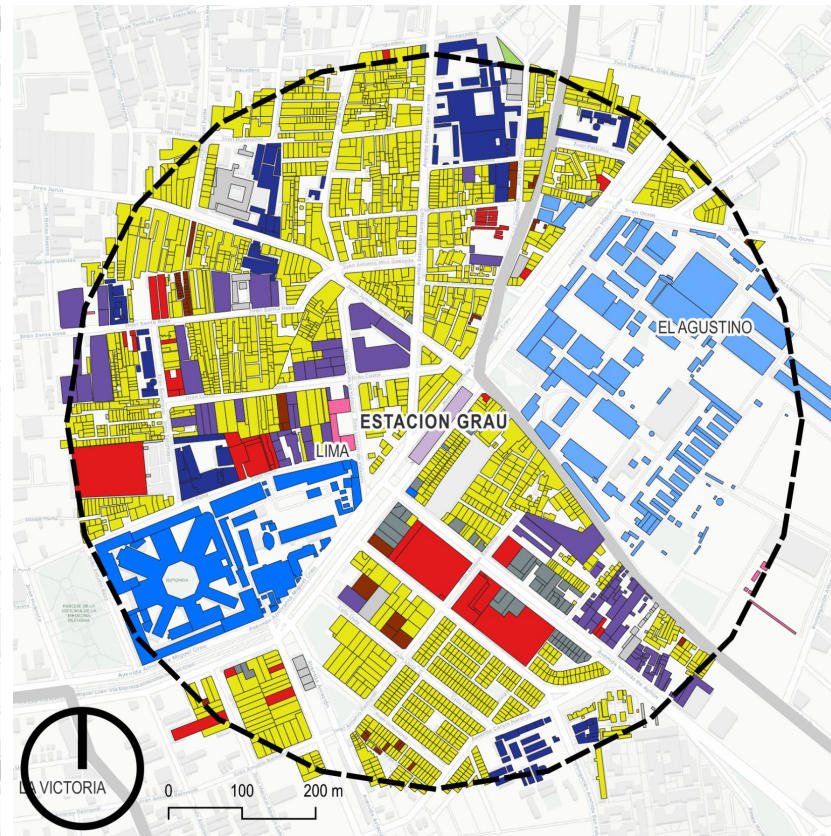


Zonin by Law map



Socioeconomic levels map

URBAN INFRASTRUCTURE



DATA GATHERING PROJECTS

Utility networks, Land Use, and
Buildings Conditions

This is a project sample. The client commissioned the collection of **+20 thematic datasets** built-up into a DB.

It was required to gather transport infrastructure, building conditions, socio-economic demographics, zoning regulation, cadastral, and utility network infrastructure; including internet accessibility data.

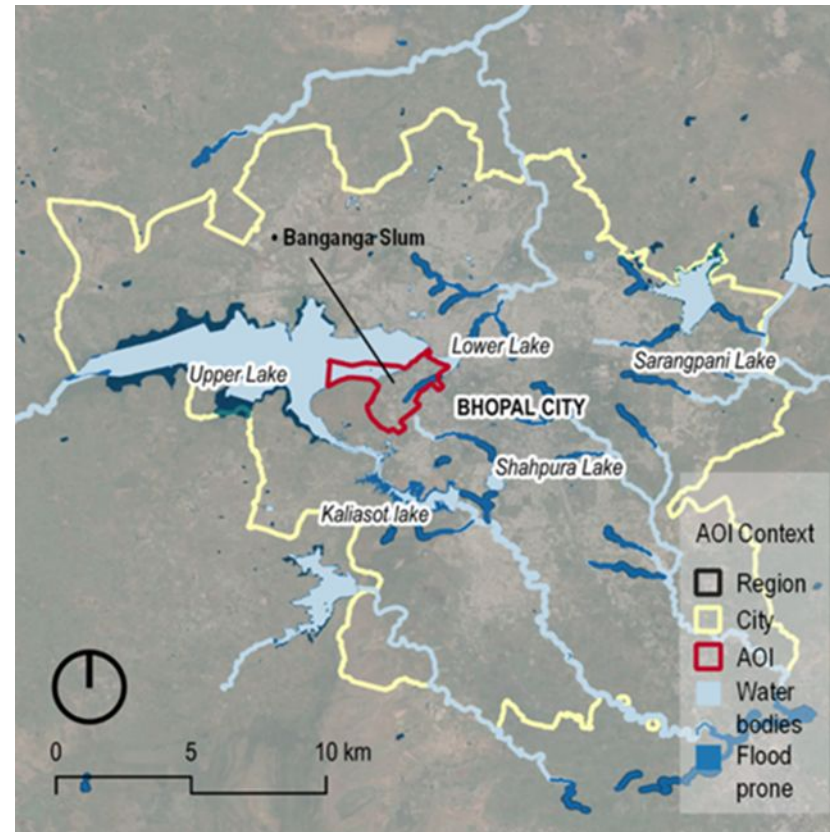
Some of the data was gathered by *accessing to REST API servers* from government entities, other data by digitizing dedicated drone surveys.

Project was delivered on time. **Data-quality surpassed external stakeholders expectations** and were satisfied with the deliverables.

URBAN DEVELOPMENT

RESEARCH PROJECTS

Data-driven Emergency Response in
Bhopal, India

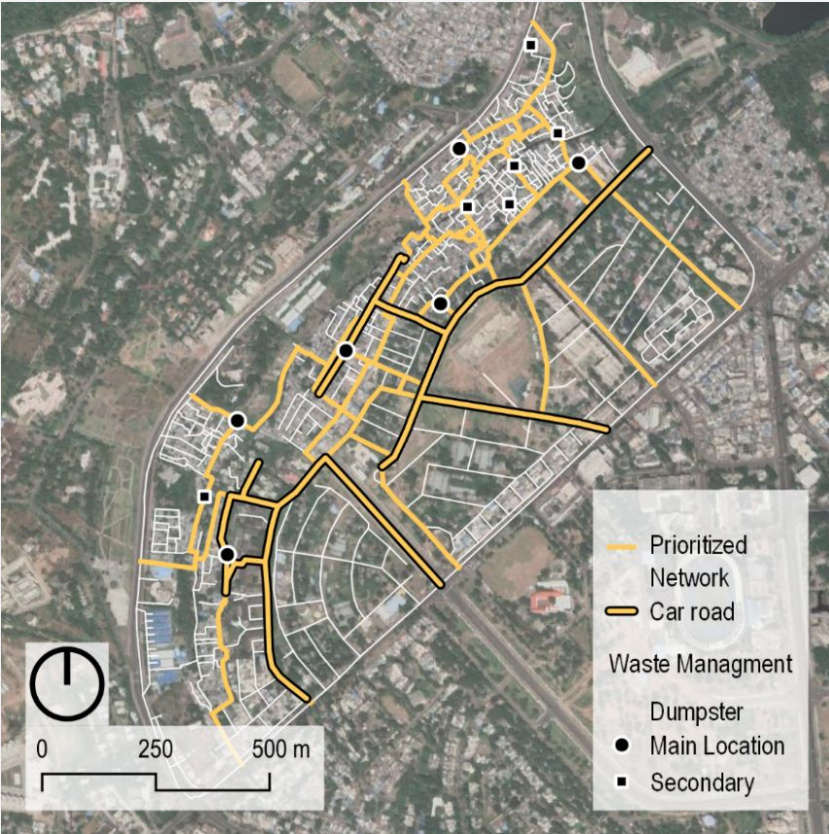
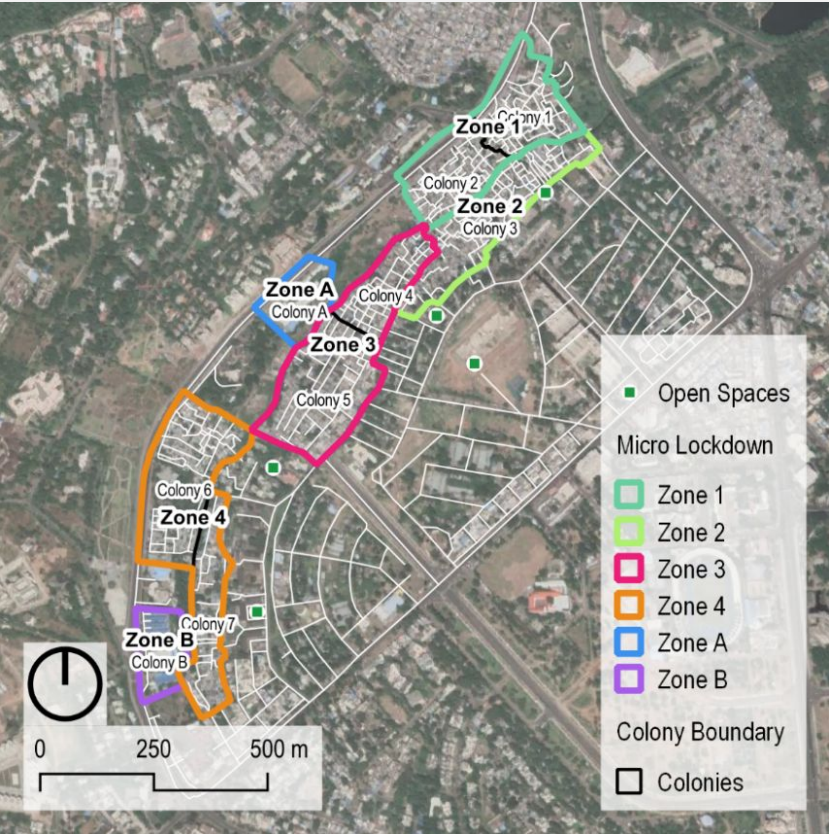


On 2021, as a part of a cross inter-institutional collaboration, GRADE (PE), IDK (UK) and Samarthan (IN) co-designed an **emergency response strategy based in spatial analysis** and participatory techniques.

The pilot project was located in *Bhopal (Madhya Pradesh, India)*, a city surrounded by water bodies.

The pilot project was tested on *Banganga*, which is a vulnerable slum, prone to flooding, with an irregular street fabric and lack of public services.

Many analysis techniques were tested at the scale of the city and the slum. A **technical report with replicability instructions** was a main project's outcome.



RESEARCH PROJECTS

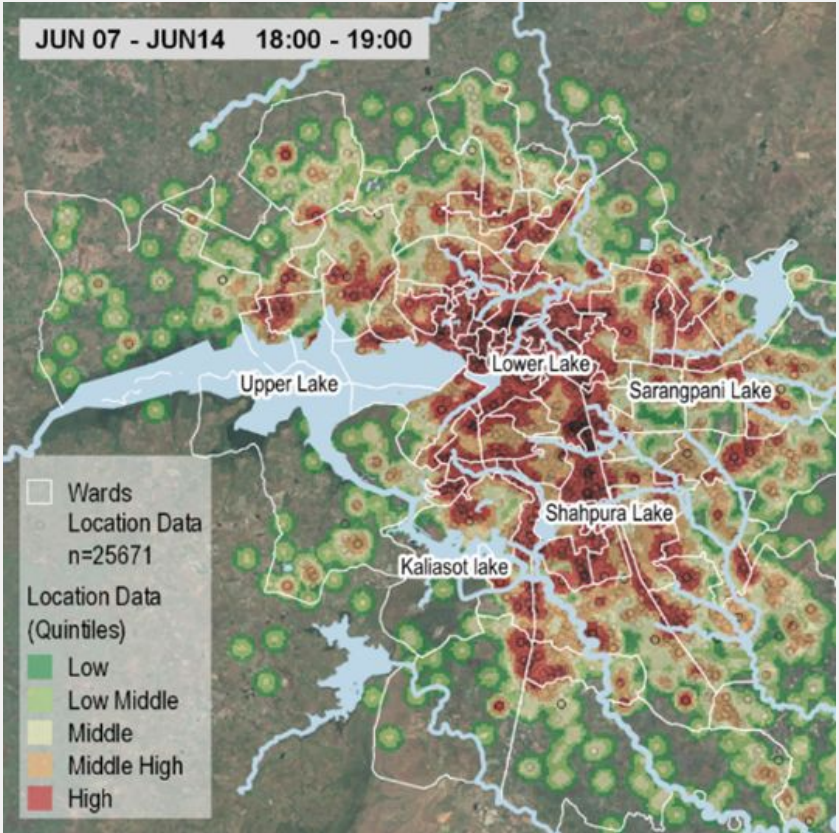
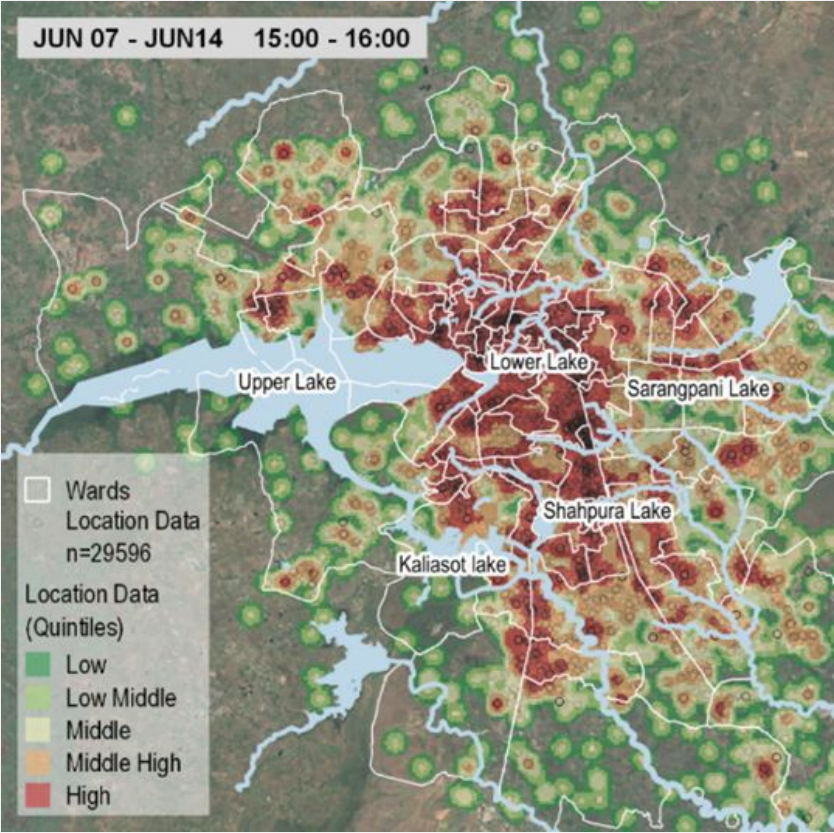
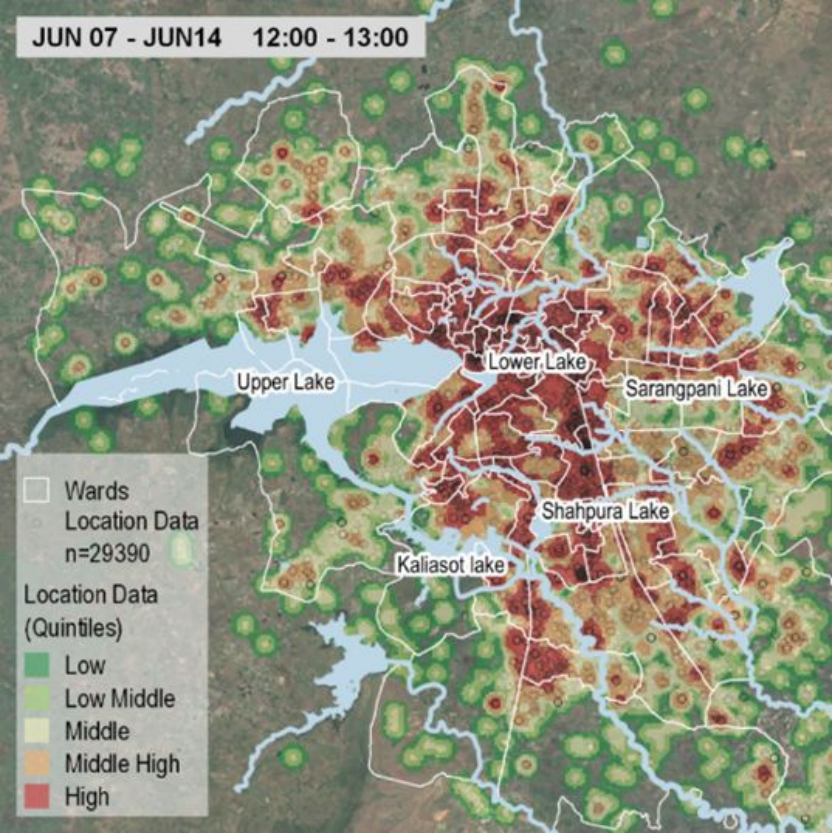
Data-driven Emergency Response in Bhopal, India

Because of the slum's shape, it wasn't clear for authorities how to reach the locals for social services.

So, a **spatial distribution analysis** was made in order to divide it in four equitable areas and two special areas. Colonies division was considered to prevent social conflicts between them.

Waste management issues were addressed by identifying the main car-able roads from the network.

Main locations inside all Micro Lockdown areas were selected to provide dumpsters. As well as secondary location according to accessibility.



One of the spatial data solutions and explorations landed on **mobile data analysis** derived from GPS signals. A sample data was tested as a source of local mobility proxy at city wide level.

Raw data was processed so that, each unique device would be aggregated into hourly and daily range maps.

The **heat maps** display the intensity of movement in the city. The data has been classified by quintiles.

It shows the congestion both in the modern and historic areas of the city. It also displays the most **active areas during the day**, providing in-detail information to identify which areas have more people consistently.

The results of this study were delivered to the local authorities of the city, to the non-profit organizations and local collaborators of the project.

This projects results were also published as well as a technical report describing all processing in order to be replicated and implemented in other areas.

THANKS!

Do you have any
questions?

Contact +1 647 787 2946 adrianchuquipiondo@gmail.com

